

PUBLIC
EXHIBIT 50

U.S. Patent No. 8,330,235 Domestic Industry Chart

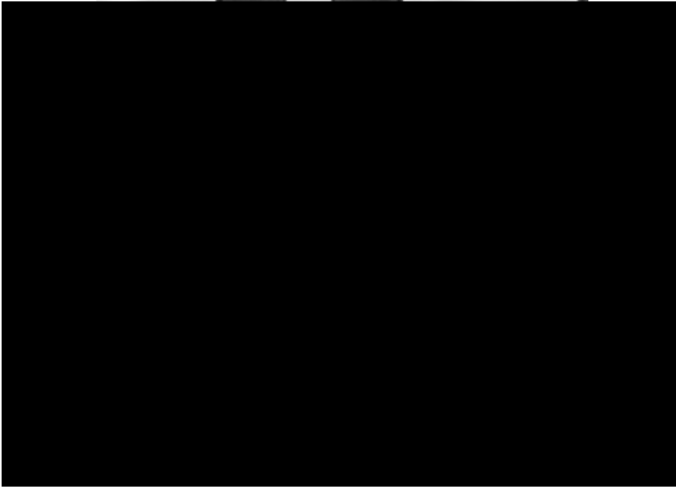
Title: Method to reduce mol damage on NiSi

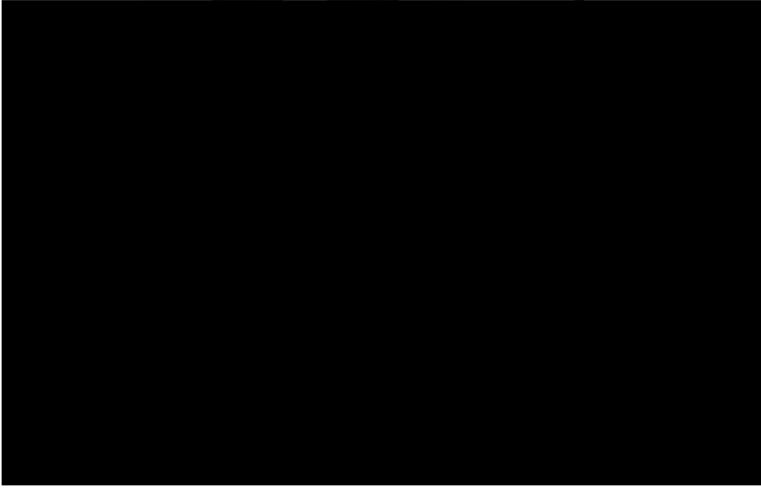
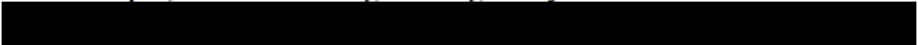
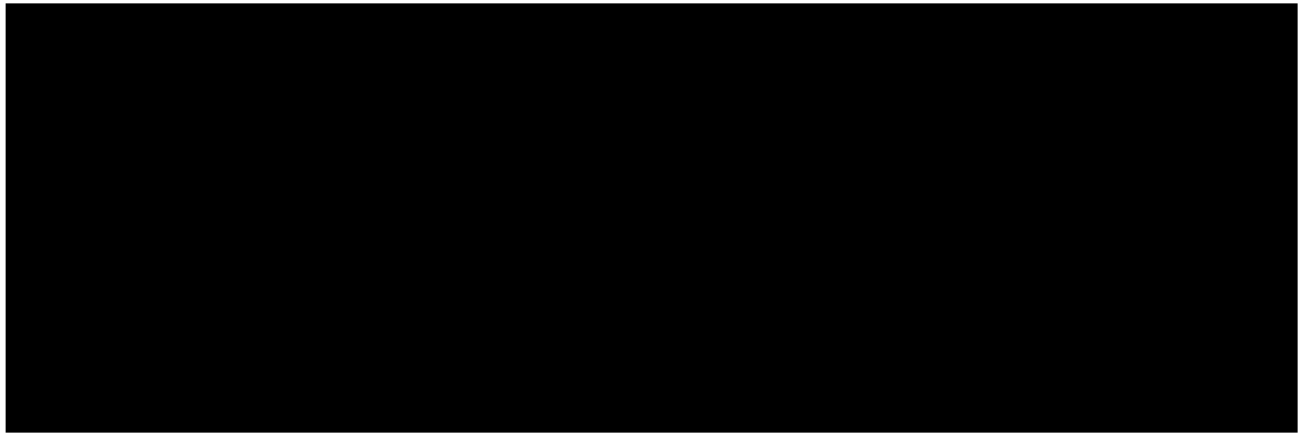
Inventors: Karthik Ramani, Paul R. Besser

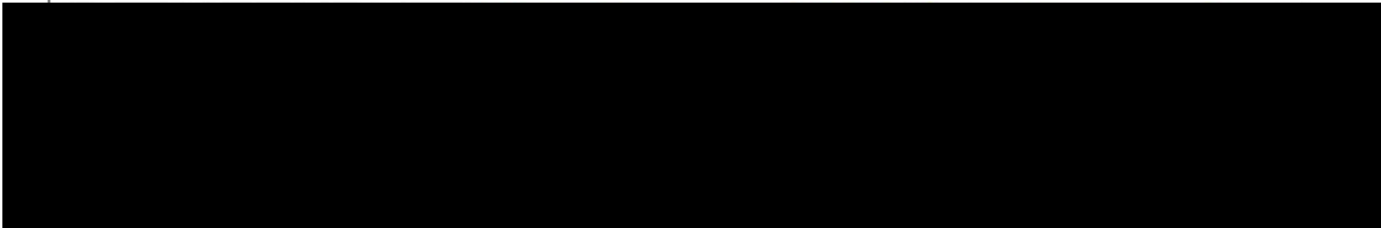
Abstract: Transistor devices are formed with nickel silicide layers formulated to prevent degradation upon removal of overlying stress liners. Embodiments include transistors with nickel silicide layers having a platinum composition gradient increasing in platinum content toward the upper surfaces thereof, i.e., increasing in platinum in a direction away from the gate electrode and source/drain regions. Embodiments include forming a first layer of nickel having a first amount of platinum and forming, on the first layer of nickel, a second layer of nickel having a second amount of platinum, the second weight percent of platinum being greater than the first weight percent. The layers of nickel are then annealed to form a nickel silicide layer having the platinum composition gradient increasing in platinum toward the upper surface. The platinum concentration gradient protects the nickel silicide layer during subsequent processing, as during etching to remove overlying stress liners, thereby avoiding a decrease in device performance.

Claim	'235 Domestic Industry Products
[1pre] A semiconductor device comprising:	To the extent the preamble is limiting, chips manufactured using GlobalFoundries' 130CBIC Process (hereinafter the "'235 Domestic Industry Products") ¹ comprise a semiconductor device, as explained further in the limitations below.
[1a] a silicon substrate,	The '235 Domestic Industry Products comprise a silicon substrate. For example, the cross-sectional TEM reverse engineering analysis below shows a semiconductor substrate (bottom).

¹ GlobalFoundries reserves the right to identify different or additional '235 Domestic Industry Products and amend or supplement these charts after further investigation and discovery, claim construction, or other meriting circumstances.

Claim	'235 Domestic Industry Products
	
[1b] a transistor, comprising source/drain regions and a gate electrode, on the substrate; and	The '235 Domestic Industry Products comprise a transistor, comprising source/drain regions and a gate electrode, on the substrate. For example, the cross-sectional reverse engineering analysis below shows a transistor with source/drain regions and a gate electrode on the substrate.

Claim	'235 Domestic Industry Products
	
[1c] a nickel silicide layer on the gate electrode and source/drain regions,	The '235 Domestic Industry Products comprise a nickel silicide layer on the gate electrode and source/drain regions. For example, the reverse engineering analysis below shows a nickel silicide layer on the gate electrode.  

Claim	'235 Domestic Industry Products
	Likewise, the reverse engineering analysis below shows a nickel silicide layer on the source/drain regions using the same color scheme. 
[1d] wherein the nickel silicide layer comprises platinum having a composition gradient, from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide, with platinum increasing in amount in a direction away from the source/drain regions and the gate electrode, from between about 8 wt. % and about 12 wt. % at the silicon interface.	In the '235 Domestic Industry Products, the nickel silicide layer comprises platinum having a composition gradient, from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide, with platinum increasing in amount in a direction away from the source/drain regions and the gate electrode, from between about 8 wt. % and about 12 wt. % at the silicon interface. For example, the reverse engineering analysis below shows a nickel silicide layer on the gate electrode. The platinum concentration in the nickel silicide layer starts between around 8 wt. % and 12 wt. % at the silicon interface and increases in an upward direction (i.e., towards an opposite surface of the nickel silicide). For example, analysis of the region in the red box in the annotated reverse engineering analysis below illustrates the composition gradient of platinum in the nickel silicide layer.

Claim	'235 Domestic Industry Products
	As shown in the material analysis chart below, the platinum concentration begins at a weight percentage between 8 wt. % and 12 wt. % at the silicon interface and increases from right to left (i.e., towards an opposite surface of the nickel silicide). The green arrow at the top of the material analysis chart below corresponds to the green arrow in the reverse engineering analysis above.

Claim	'235 Domestic Industry Products
	The material analysis chart below shows a magnified version of the graph above along the y-axis. As shown, the platinum concentration begins at a weight percentage between about 8 wt. % and 12 wt. % at the silicon interface and increases moving from right to left (i.e., towards an opposite surface of the nickel silicide).

Claim	'235 Domestic Industry Products